Hypertension Research
https://doi.org/10.1038/s41440-026-02768-5

REVIEW ARTICLE

Magnesium and blood pressure regulation: systemic and renal mechanisms

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Received: 23 May 2026 / Revised: 16 July 2026 / Accepted: 25 July 2026
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Abstract

Experimental and clinical evidence have demonstrated that Mg deficiency contributes to elevated blood pressure (BP), whereas Mg supplementation exerts modest but consistent antihypertensive effects. The systemic effects of Mg supplementation and/or deficiency on BP have been investigated in relation to vascular smooth muscle cells (VSMCs), the renin–angiotensin–aldosterone system (RAAS), endothelial function, the sympathetic nervous system, and inflammation associated with oxidative stress. Some of these mechanisms are supported by compelling evidence, whereas others remain controversial. In the kidney, following glomerular filtration, ~95% of filtered Mg is reabsorbed along the renal tubules. In the proximal tubule (PT), Mg reabsorption occurs mainly via a paracellular transport pathway driven by a negative transepithelial electrical gradient. In contrast, in the thick ascending limb of Henle's loop (TAL), Mg reabsorption is regulated by electrochemical gradients generated through the coordinated activity of transporters such as the Na-K-2Cl cotransporter (NKCC2) and Na-K-ATPase. In the distal convoluted tubule (DCT), transient receptor potential melastatin 6 and 7 (TRPM6/7) channels tightly regulate the transcellular Mg reabsorption pathway. Hypertension is associated with increased urinary Mg excretion, resulting in relative Mg deficiency. Accumulating evidence suggests that Mg deficiency contributes to the pathogenesis of salt-sensitive and/or implementation hypertension, presumably through mechanisms involving Mg, Na and K transporters, mainly in the TAL and DCT.In this review, we discuss both systemic and renal mechanisms linking Mg to BP regulation. A better understanding of the interactions among Mg, Na, and K transport may provide novel insights into the role of Mg in hypertension.

Keywords Magnesium · Implementation hypertension · Renal Mg transport · Salt sensitivity

Introduction

The pathogenesis of essential hypertension is closely associated with salt sensitivity, and its underlying mechanisms have been investigated primarily in relation to sodium (Na) retention and its interaction with potassium (K). Calcium (Ca) has also been implicated in the pathogenesis of hypertension through its role in intracellular regulation of vascular contractility. Magnesium (Mg), by contrast, is involved in the activation of more than 300 enzymatic cofactors and plays essential roles in muscle contraction, neurotransmission, inflammation, oxidative stress, body temperature and blood pressure (BP). However, despite its physiological importance, the detailed mechanisms by which Mg participates in BP regulation have not been fully elucidated, and as a result it has long been referred to as "the forgotten ion in the pathogenesis of hypertension".

The human body contains ~20–28 g of Mg, of which 50–60% is stored in bone and teeth, 34–39% in soft tissues such as skeletal muscle and visceral organs, and only 1–2% in blood and extracellular fluid. Serum Mg exists in three forms: 5–15% is complexed with phosphate or carbonate, 20–30% is protein-bound, and the remaining 55–70% exists as ionized Mg. The serum Mg concentration is normally 0.7–1.0 mmol/L. Because only approximately 1% of total body Mg is present in the extracellular milieu, even a slight reduction in serum Mg concentration may reflect substantial depletion of total body Mg stores.

Mg deficiency has been associated with hypertension, diabetes mellitus, osteoporosis, arrhythmias, ischemic heart disease, and metabolic syndrome. Early observational studies reported inverse associations between serum Mg concentrations and the prevalence of hypertension, cardiovascular risk, and mortality [1]. Subsequently,

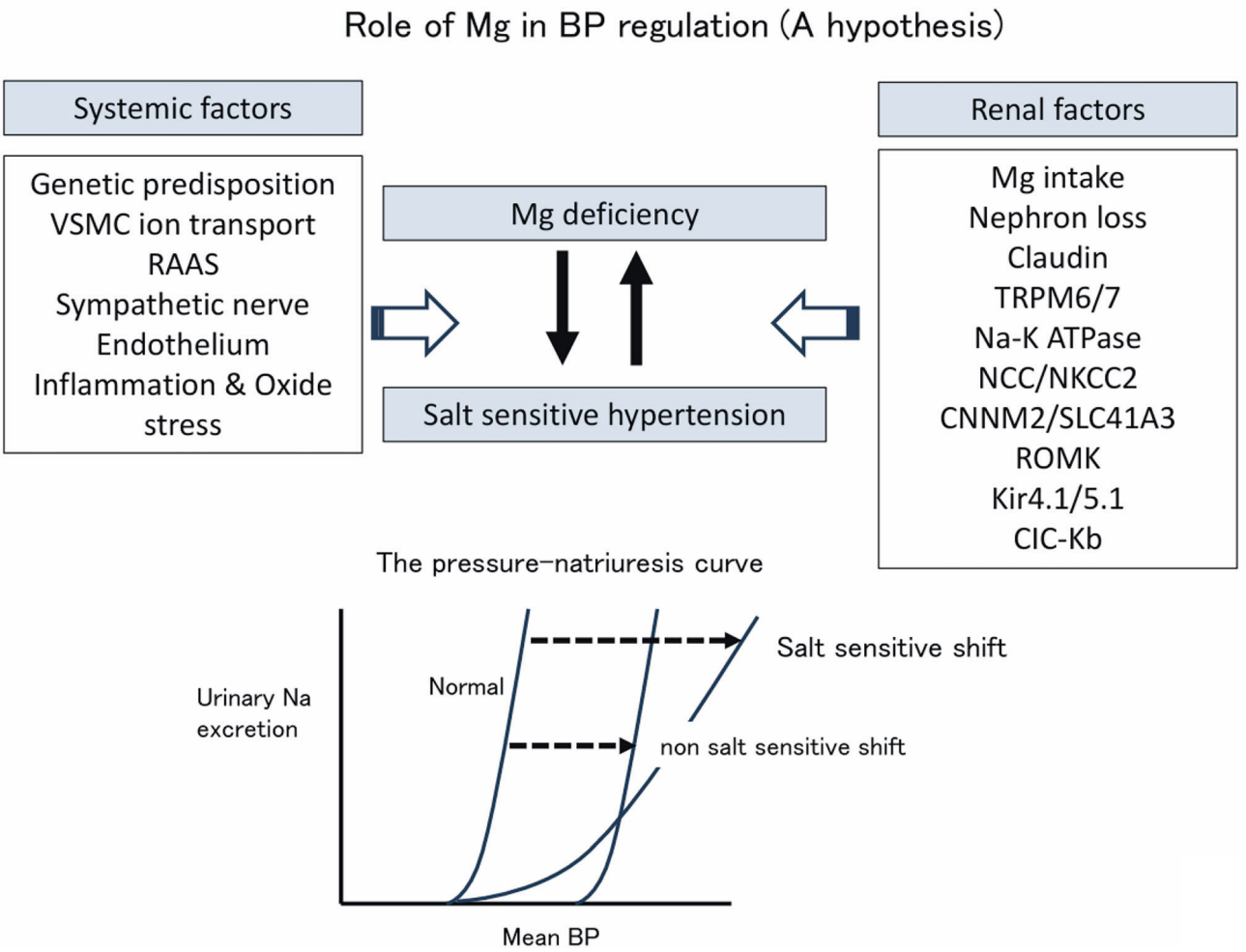
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Published online: 04 August 2026

Graphical Abstract

Mg exerts both BP raising and lowering effects depending on the body Mg status. Mg deficiency is associated with increase in BP through the multifaceted systemic and renal mechanisms. Experimental studies have shown that Mg deficiency–induced hypertension can be salt-sensitive and may be associated with implementation hypertension. The pressure-natriuresis curve shifts rightward and downward as salt sensitivity develops, requiring a higher BP to achieve the same level of sodium excretion



numerous randomized controlled trials (RCTs) and meta-analyses have confirmed that Mg supplementation exerts modest but consistent antihypertensive effects.

Meanwhile, recent advances in renal physiology have demonstrated that renal Mg transport is closely integrated with Na and K handling through mechanisms involving Na-K-2Cl cotransporter (NKCC2) and Na-K-ATPase, and is coordinately regulated with Mg channels transient receptor potential melastatin 6 and 7 (TRPM6/7). In addition, Mg deficiency has been implicated in pathophysiological conditions such as salt-sensitive hypertension, diabetes mellitus, and metabolic syndrome, increasing interest in its potential as a therapeutic target.

Based on the above-mentioned background, this review focuses on the relationship between Mg and BP regulation. We summarize the available clinical evidence and discuss its pathophysiological significance from both systemic and renal mechanistic perspectives.

Experimental and clinical evidence linking Mg to BP regulation

The relationship between Mg and BP has been recognized since the 1920s, when case reports suggested that Mg supplementation was associated with BP reduction in

patients with acute nephritis complicated by hypertension [2]. Subsequent experimental studies demonstrated that Mg supplementation exerts antihypertensive effects in spontaneously hypertensive rats (SHR) and deoxycorticosterone acetate (DOCA)-salt hypertensive rats [3 4]. Shimosawa et al. reported that intravenous administration of Mg in Sprague–Dawley rats attenuated the pressor response to sympathetic nerve stimulation, suggesting sympathetic inhibition as one of the antihypertensive mechanisms of Mg [5]. In addition, Atarashi et al. suggested that Mg infusion may suppress angiotensin II (AII)-mediated pressor responses [6], implicating the suppression of renin–angiotensin–aldosterone system (RAAS) in the antihypertensive actions of Mg. Conversely, numerous studies have shown that Mg deficiency is associated with elevated BP [7–9]. Kumagai et al. interestingly suggested that one of the mechanisms underlying this effect is the induction of salt-sensitive hypertension by Mg-deficient diets [10].

Since the 1990s, many RCTs and clinical trials have reported antihypertensive effects of Mg supplementation. In a meta-analysis of 34 RCTs involving more than 2000 participants, Zhang et al. demonstrated that Mg supplementation reduced systolic blood pressure (SBP) by 2 mmHg and diastolic blood pressure (DBP) by 1.78 mmHg [11]. Similar findings have been consistently reproduced in multiple subsequent meta-analyses [12–15] (Table 1). Collectively, these data provide clear evidence that Mg supplementation induces modest but statistically significant reductions in BP. These clinical trials have also identified several important clinical characteristics of Mg responsiveness:

1. The BP-lowering effect of Mg is dose-dependent, with efficacy generally observed at daily doses of ~240–600 mg;
2. The effect is more pronounced in individuals with Mg deficiency;
3. Responsiveness is greater in patients with uncontrolled hypertension, regardless of antihypertensive treatment status, as well as in untreated hypertensive patients;
4. Mg supplementation does not lower BP in normotensive individuals;
5. The antihypertensive effect appears to be greater in patients with diabetes mellitus.

These clinical findings suggest that Mg deficiency contributes to BP elevation and that Mg supplementation has antihypertensive properties. Importantly, these effects may not be uniformly applicable across the entire hypertensive population, but rather may be more effective in specific clinical phenotypes.

Systemic mechanisms linking Mg to BP regulation

The systemic effects of Mg on BP, including both antihypertensive and pressor modulation, are mediated through multiple physiological mechanisms [16]. The integrated actions of the following mechanisms collectively contribute to BP regulation.

Regulation via vascular smooth muscle cells (VSMC)

Mg suppresses Ca influx through intracellular voltage-dependent L-type Ca channels, thereby reducing VSMC contractility [17]. Mg deficiency increases intracellular Ca concentrations and promotes vasoconstriction. In addition, intracellular Mg suppresses Ca release from the sarcoplasmic reticulum (SR), thereby inhibiting muscle contraction [18]. Mg also facilitates the activity of SR Ca-ATPase, promoting the reuptake of Ca released from the SR back into intracellular storage sites [19]. Taken together, Mg deficiency enhances Ca release from the SR and augments VSMC contractility, thereby contributing to both the development and maintenance of hypertension [20].

Regulation via endothelial function

In cultured endothelial cells, Mg enhances endothelial nitric oxide synthase (eNOS) activity and nitric oxide (NO) production [21]. In murine models, Mg-induced arterial vasodilation is mediated by endothelium-dependent NO signaling, and this response is attenuated by inhibition of eNOS with Nω-nitro-L-arginine methyl ester (L-NAME) [22]. Clinical studies in patients with coronary artery disease have demonstrated that Mg supplementation improves brachial artery flow-mediated dilation (FMD), an NO-dependent vasodilatory response, as well as exercise tolerance [23]. Mg deficiency is associated with endothelial dysfunction, increased oxidative stress, and impaired vasodilatory capacity. Furthermore, Mg promotes prostacyclin production and suppresses endothelin-1 activity, thereby exerting antihypertensive effects.

Regulation via the sympathetic nervous system

Mg deficiency enhances sympathetic nervous system activity through multiple mechanisms, including increased N-methyl-D-aspartate (NMDA) receptor activity and augmented catecholamine release [24]. Indeed, Mg supplementation has been shown to modulate catecholamine release and suppress N-type Ca channels at sympathetic nerve terminals, thereby reducing sympathetic tone [5]. In

Table 1 The effect of Mg supplementation on BP (clinical trials)

Trials included (129 in total)	*N*	Mg dose (mg/day)	Reduction in SBP/DBP or Trials of BP lowering/Total Trials	Reference
34	2028	368 mg(average)	SBP: −2 mmHg DBP: −1.78 mmHg	Zhang et al. [11]
11	538	365–450 mg	SBP: −4.18 mmHg DBP: −2.27 mmHg	Dibaba et al. [12]
28	2709	82–637 mg 365 mg(average)	SBP: −2.8 mmHg DBP: −2.05 mmHg (Effective in hypomagnesemia)	Argeros et al. [13]
49	NA	250–632 mg 304–538 mg 120–972 mg 240–607 mg	Normotensives: 0/17 Well-controlled: 0/1 Untreated: 5/21 Poorly-controlled: 10/10	Rosanoff et al. [14]
7 (T2DM)	333	Over 300 mg	SBP: −5.78 mmHg DBP: −2.5 mmHg	Asbaghi et al. [15]

A total of 129 clinical trials included in major meta-analyses were summarized. Across these analyses, Mg supplementation at doses ranging from ~300 to 600 mg/day consistently demonstrated antihypertensive effects. Notably, the BP-lowering effect appeared to be greater in patients with diabetes mellitus, particularly those with T2DM. In contrast, no significant BP-lowering effect was found in either normotensive healthy individuals or well-controlled group. Partially adapted from refs. [11–15]

NA not available, *T2DM* Type 2 diabetes mellitus

humans, BP elevations induced by catecholamine release during endotracheal intubation have been shown to be attenuated by intravenous Mg administration [25].

Regulation via the renin–angiotensin–aldosterone system (RAAS)

Whether Mg directly modulates BP through the RAAS remains controversial, with both supportive and conflicting evidence reported. Animal studies have demonstrated aldosterone levels remained the same under Mg-deficient dietary conditions [26]. However, Laurant et al. observed increase in aldosterone in rats given Mg-deficient diet [27]. In human, Corica et al. reported that intravenous Mg administration in healthy adults reduced serum aldosterone concentrations, and because atrial natriuretic peptide (ANP) levels remained unchanged, this effect was unlikely to be secondary to extracellular volume expansion [28]. Studies of oral Mg supplementation in humans have yielded inconsistent findings. Some reported no reductions in either BP or serum aldosterone concentrations [29 30], whereas others observed BP reductions without significant changes in aldosterone levels [31 32]. Data regarding upstream RAAS components are even more complex. In rat models, some studies found no change in angiotensin II (AII) levels during Mg deficiency [33], whereas others reported increases in both AII and plasma renin activity (PRA) [27]. In MRI-based murine studies evaluating body fluid composition, Mg-deficient diets increased BP without altering body fluid distribution [34]. Taken together, these findings suggest that systemic Mg deficiency may contribute to elevated aldosterone concentrations to some extent, whereas upstream RAAS components such as PRA and AII appear to remain unchanged.

Regulation via inflammation and oxidative stress

Inflammatory mechanisms induced by Na loading are increasingly recognized as contributors to hypertension, and these pathways appear to be modulated by Mg. At least two inflammatory mechanisms are implicated in salt-induced BP elevation. First, high-salt diets increase oxidative stress in antigen-presenting cells (APCs), thereby promoting hypertensive responses [35]. Second, high-salt intake upregulates the expression of NLR family pyrin domain-containing 3 (NLRP3) inflammasome components, particularly in the renal medulla [36]. Activation of NLRP3 catalyzes the production and secretion of proinflammatory cytokines such as interleukin (IL)-1β and IL-18 [37]. Genetic or pharmacological inhibition of NLRP3 suppresses the development of hypertension in murine models, suggesting a protective effect [38]. Accumulating evidence also supports an immunomodulatory role of Mg in hypertension. Mg-deficient diets stimulate proinflammatory pathways that promote BP elevation. In experimental animals, Mg deficiency induces leukocytosis and increases circulating inflammatory cytokines, including IL-1β [39]. Conversely, oral Mg supplementation reduces circulating C-reactive protein (CRP) levels [40]. These findings suggest that Mg deficiency promotes inflammatory cytokine production, oxidative stress, and endothelial injury. The relationship between Mg deficiency and immune-mediated hypertension represents an emerging area of investigation. Future studies should determine whether Mg deficiency directly induces

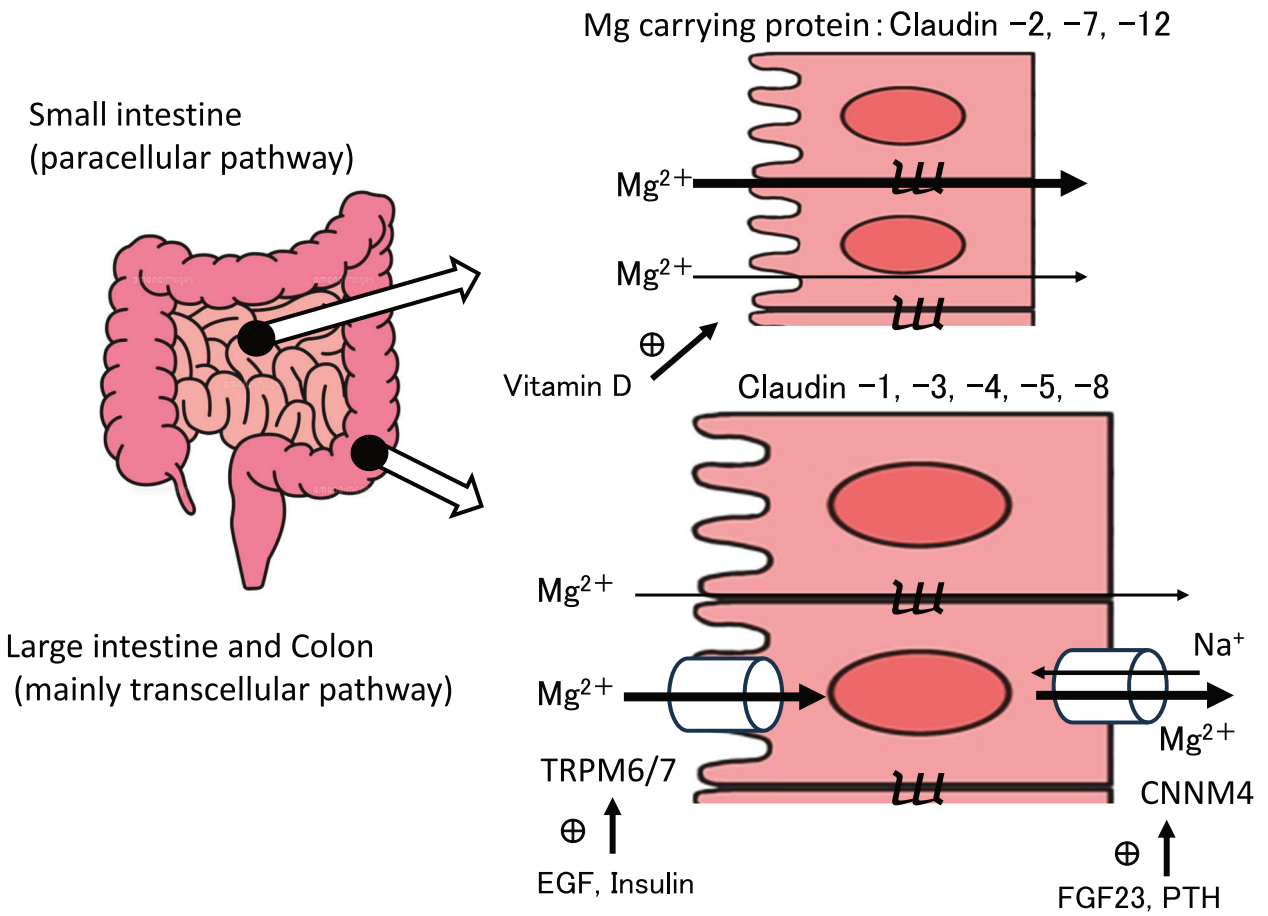


Fig. 1 Mg Transporters in the Intestine (Absorption Pathway). The mechanisms of intestinal Mg absorption differ somewhat between the small and large intestines. In the small intestine, ~90% of Mg absorption occurs via the paracellular pathway, whereas ~10% occurs via the transcellular pathway. The driving force for this transport is provided by transepithelial electrochemical gradients. Claudin-2, -7, and -12, transmembrane proteins localized at epithelial tight junctions, mediate paracellular Mg transport. The negative electrical potential generated by these proteins (~−1.2 mV) facilitates Mg absorption from the intestinal lumen. In contrast, Mg absorption in the large intestine depends primarily on transcellular transport. TRPM6/7 channels located on the luminal membrane mediate Mg uptake into epithelial cells, whereas cyclin M4 (CNNM4), located on the basolateral membrane, mediates Na–Mg exchange and transports Mg into the circulation. CNNM4 is also a molecular target of fibroblast growth factor 23 (FGF23) and parathyroid hormone (PTH), both of which suppress intestinal Mg absorption. Vitamin D enhances Ca absorption in the small intestine and concomitantly increases Mg absorption, thereby promoting a positive Mg balance. Adapted from refs. [16 41]

inflammation leading to hypertension in humans, and whether Mg supplementation can ameliorate salt-induced inflammatory hypertension [16].

Transporters linking Mg to BP regulation

Intestinal Mg absorption

Following oral intake, ~30–50% of ingested Mg is absorbed through the intestine under normal physiological conditions (Fig. 1). This absorption rate increases to as high as 80% during Mg deficiency. In the small intestine, the majority of Mg absorption occurs via the paracellular pathway via Mg-carrying protein claudin-2, -7, and -12, localized within tight junctions. In contrast, Mg absorption in the colon occurs predominantly via transcellular transport. TRPM6/7 located on the luminal membrane facilitate cellular Mg uptake, whereas CNNM4 on the basolateral membrane mediates Na–Mg exchange, transporting Mg into the circulation and thereby contributing to positive Mg balance [41].

Renal Mg excretion

The kidney is the principal route of Mg excretion and serves as the central organ responsible for Mg homeostasis, linking Mg metabolism to BP regulation (Fig. 2). Approximately 2400 mg of Mg is filtered daily by the glomeruli, of which ~95% is reabsorbed along the nephron. The proportion and mechanisms of Mg transport differ among the PT, TAL, and DCT.

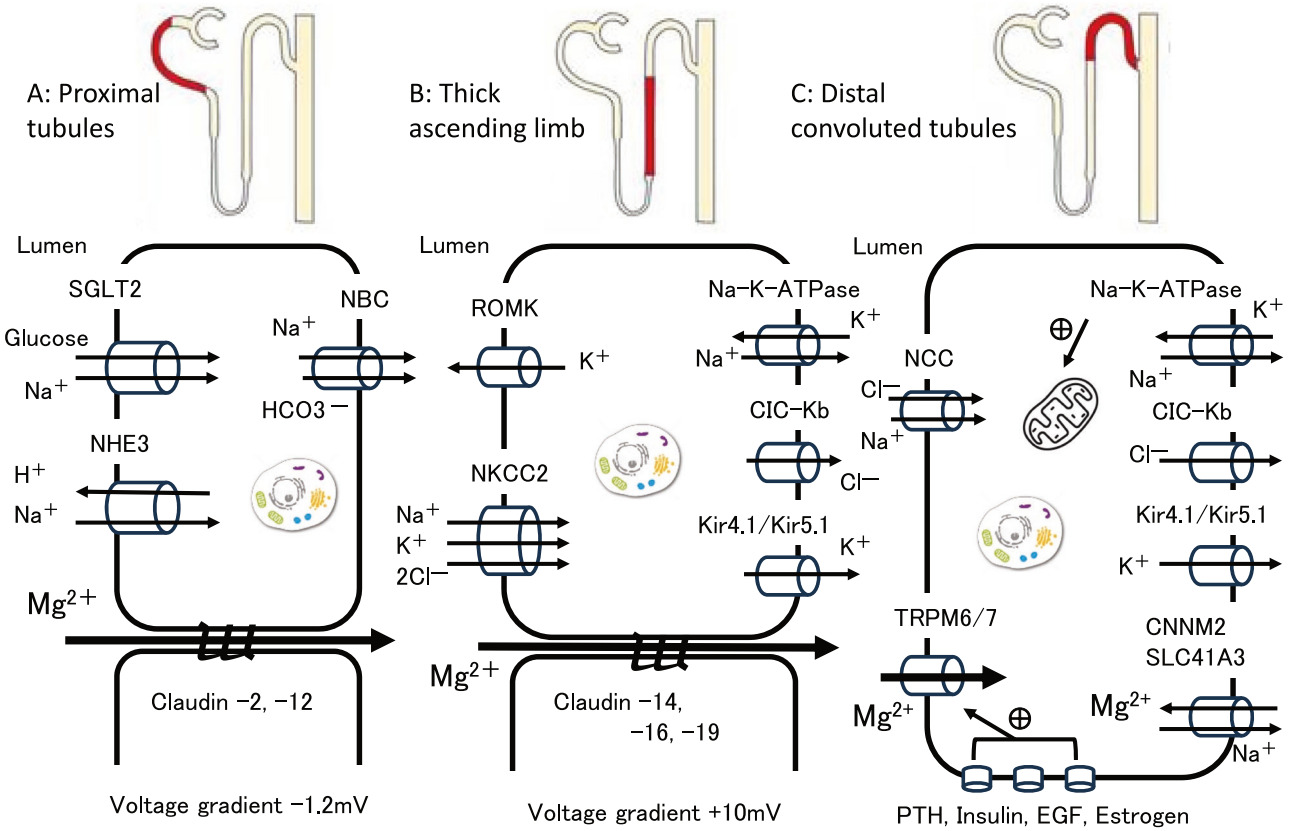


Fig. 2 Ion Transporters in the kidney (Mg excretion pathways). In the PT, ~20–30% of filtered Mg is reabsorbed. Mg reabsorption in this segment is driven by a negative transepithelial voltage gradient (~ −1.2 mV) generated by Claudin-2 and Claudin-12 within tight junctions, thereby promoting paracellular Mg transport. However, Mg transport in the PT has relatively limited regulatory significance, because downstream nephron segments, particularly the TAL and DCT, can compensate by modulating Mg transport. In addition to the above-mentioned Mg transporters, Na transporters such as sodium–glucose cotransporter 2 (SGLT2), Na–H exchanger 3 (NHE3), and Na–HCO3 cotransporter (NBC) are abundantly expressed in the PT. The TAL accounts for 50–70% of Mg reabsorption. Here, Mg reabsorption is driven by the lumen-positive voltage gradient (~ +10 mV) generated by the coordinated activity of the Na-K-2Cl cotransporter (NKCC2) (furosemide-sensitive), Na-K-ATPase, renal outer medullary potassium channel (ROMK), inwardly rectifying potassium channels (Kir4.1/5.1), and voltage-dependent chloride channel ClC-Kb. This electrochemical gradient promotes paracellular Mg reabsorption. In addition, Mg serves as a cofactor for Na-K-ATPase, contributing directly to ATPase activity in the TAL. The DCT accounts for 5–10% of Mg reabsorption. In this segment, Mg is actively reabsorbed via transcellular transport mediated by transient receptor potential melastatin6 and 7 (TRPM6/7) Mg channels. Unlike Mg reabsorption in the PT, Mg transport in the DCT is tightly regulated and is considered the principal determinant of final urinary Mg excretion. The DCT also plays an important role in Na balance, and transport systems such as cyclin M2 (CNNM2) and solute carrier family 41 member 3 (SLC41A3) are thought to link Mg metabolism to Na handling, thereby contributing to salt-sensitive hypertension. Furthermore, receptors for PTH, insulin, and epidermal growth factor (EGF) are expressed in the DCT and stimulate TRPM6/7 activity [16]. For example, although PTH is primarily recognized as a hormone that enhances Ca reabsorption in the DCT and TAL, it also concomitantly promotes Mg reabsorption through TRPM6/7 and paracellular pathways, thereby shifting Mg balance in a positive direction. Na-Cl cotransport (NCC) (thiazide-sensitive) reabsorbs luminal Na into the DCT cells. Conversely, PTH deficiency is often associated with hypocalcemia and hypomagnesemia. Partially adapted from refs. [16 41]

Mg transporters in the PT

Approximately 20–30% of filtered Mg is reabsorbed in the PT. The driving force for Mg reabsorption is the negative transepithelial voltage gradient generated by Claudin-2 and Claudin-12 within tight junctions, enhancing paracellular Mg reabsorption. Although Na transporters such as SGLT2, NHE3 and NBC are abundant in the PT, Mg transport is not directly regulated by these transporters. Instead, Mg handling related to Na transport is predominantly regulated downstream in the TAL and DCT through transporters such as NKCC2, Na-K-ATPase and the others (Fig. 2) [41].

Mg transporters in the TAL

The TAL accounts for 50–70% of total Mg reabsorption. Mg reabsorption occurs through both paracellular and transcellular pathways. As for the former, Claudin-14, -16, and -19

localized at intercellular junctions regulate Mg transport. Among these, Claudin-16 and Claudin-19 facilitate Mg transport, whereas Claudin-14 appears to exert inhibitory effects. The TAL also contains Na-K-ATPase, ROMK, and NKCC2, which also contribute to the generation of a lumen-positive transepithelial voltage gradient. This electrochemical voltage gradient serves as the driving force for Mg reabsorption from the tubular lumen (Fig. 2) [41].

Mg transporters in the DCT

The DCT reabsorbs ~5–10% of filtered Mg. Here, Mg is actively reabsorbed through transcellular transport mediated by the Mg channels, TRPM6 and TRPM7. Unlike Mg reabsorption in the PT, Mg transport in the DCT is tightly regulated and, together with the TAL, determines final urinary Mg excretion. The DCT is also critically involved in Na balance and represents an important site linking Mg and Na metabolism through other transporters such as CNNM2 and SLC41A3, thereby contributing to salt-sensitive hypertension. Receptors for parathyroid hormone (PTH), insulin, and epidermal growth factor (EGF) are also expressed in the DCT, and these hormones regulate Mg transport. For example, TRPM6/7 activity is stimulated by insulin, PTH, EGF, and estrogen [16 41] (Fig. 2). Notably, meta-analyses have shown that sodium–glucose cotransporter 2 (SGLT2) inhibitors increase serum Mg concentrations [42–45]. This effect is thought to result from inhibition of Na reabsorption in the PT, leading to enhanced NKCC2 activity in the downstream TAL. Consequently, the lumen-positive voltage gradient is augmented, increasing paracellular Mg reabsorption and thereby elevating serum Mg levels. Improvement in hyperinsulinemia induced by SGLT2 inhibitors may also enhance Mg reabsorption and reduce fractional Mg excretion, contributing further to increased serum Mg concentrations.

Genetic determinants of Mg metabolism

Since 2000, advances in DNA sequencing technologies have facilitated genome-wide association studies (GWAS), broadening genetic investigations from monogenic disorders to common diseases. Several genetic abnormalities underlying inherited hypomagnesemia have been identified. In the TAL, mutations in CLDN16, CLDN19, RRAGD, and CARS impair paracellular Mg transport and cause hypomagnesemia. In the DCT, mutations in TRPM6 or TRPM7 result in hypomagnesemia, often accompanied by hypocalcemia. In addition, hereditary tubulopathies such as Gitelman syndrome are characterized by impaired Na reabsorption and are associated with hypomagnesemia, hypokalemia, and metabolic alkalosis due to mutations in genes including SLC12A3, CLCNKB, KCNJ10, KCNJ16, ATP1A1, FXYD2, HNF1B, and PCBD1 [41]. For a comprehensive review of inherited hypomagnesemia, the review by de Baaij et al. is highly recommended [46]. Regarding its pathophysiological role, although patients with inherited hypomagnesemia tend to exhibit hypotension, little is known about the cause-and-effect relationship between genetic abnormalities of Mg and BP regulation.

Mg transport and salt-sensitive hypertension

Excessive dietary salt intake induces hypertension, primarily through extracellular volume expansion resulting from Na retention. During salt loading, urinary Mg excretion increases, leading to a negative Mg balance. This is thought to occur because salt loading impairs NKCC2 in the TAL, thereby attenuating the lumen-positive voltage gradient that serves as the driving force for Mg reabsorption. In addition, high-salt conditions have been reported to downregulate the expression of the Mg channel TRPM6 in the DCT, further increasing urinary Mg excretion. This interaction may represent one of the mechanistic links between excessive salt intake, salt-sensitive hypertension, and Mg deficiency. Animal studies suggest that Mg supplementation may ameliorate salt-sensitive hypertension through modulation of both Na and K homeostasis. Ploth et al. demonstrated in rats that intravenous Mg reduces Na reabsorption in the TAL [47]. This effect is mediated through activation of the calcium-sensing receptor (CaSR) at this nephron segment and resembles the natriuretic action of loop diuretics through inhibition of NKCC2. Mg is also closely linked to K homeostasis. Potassium supplementation is known to exert antihypertensive effects in patients with hypertension. Although multiple mechanisms have been proposed, one contributing factor is the observation that K deficiency stimulates NHE3 activity in the proximal tubule [48]. Terker et al. further reported that K-deficient diets stimulate the NCC in the DCT, thereby promoting Na retention [49]. Moreover, Shils et al. observed in humans that Mg deficiency itself can induce K deficiency [50]. This is thought to result from enhanced urinary K excretion, likely mediated by increased ROMK activity and altered aldosterone-dependent K handling [51]. Collectively, these findings suggest that Mg deficiency promotes Na retention through enhanced NCC activity in the DCT. This supports the concept of a "renal Mg-Na-K axis," whereby high salt intake disrupts TRPM6-mediated Mg regulation in the DCT, leading to increased Mg loss, dysregulation of Na transport, and a rightward shift of the Guyton pressure–natriuresis curve, thereby exacerbating salt-sensitive hypertension. Of particular interest, Funato et al. reported that DCT-specific knockout of TRPM6 in mice suppressed renin secretion from the adjacent juxtaglomerular apparatus (JGA) and abolished circadian BP

variation [52], suggesting a potential role for this mechanism in the development of human "Implementation hypertension" [53]. All in all, these findings suggest the existence of a novel hypothetic mechanism linking Mg homeostasis to BP regulation. Sufficient Mg supplementation may conversely ameliorate hypertension. This newly recognized interaction may provide a novel pathogenic framework for understanding Mg and salt sensitivity.

Does Mg-induced BP reduction improve cardiovascular outcomes?

In recent years, health-oriented online information has increasingly promoted Mg supplementation through dietary guidance and supplements. An important clinical question is whether Mg supplementation truly improves hard clinical endpoints, including death, major adverse cardiovascular events (MACE) and/or primary renal outcomes such as delaying the progression into end-stage renal failure, and ultimately contributes to improved patients' prognosis. In routine clinical practice, however, BP reduction should not be equated with improved cardiovascular outcomes. At present, it is more appropriate to conclude that although Mg supplementation exerts modest antihypertensive effects, there is insufficient evidence to state that it improves cardiovascular outcomes. This is because no large-scale RCTs have yet demonstrated the cause-and-effect relationships between Mg supplementation and reductions in MACE and/or primary renal endpoints. Dietary Mg interventions likely produce more gradual BP reductions than pharmacological antihypertensive therapies and are often evaluated in generally healthy populations, making outcome detection more difficult in RCT.

Mg supplementation for health maintenance and quality of life

Mg intake in healthy individuals

Dietary Mg intake in the Japanese population has consistently remained below recommended levels. According to the National Health and Nutrition Survey conducted by the Ministry of Health, Labour and Welfare, average Mg intake falls below recommended dietary allowances across most age groups. For example, in the 2023 survey, average daily Mg intake among men aged 20–49 years was ~229–241 mg/day, whereas that among women was ~193–214 mg/day. In contrast, the Dietary Reference Intakes for Japanese (2025 edition) recommend ~330–380 mg/day for adult men and 280–290 mg/day for adult women [54]. These data suggest that many Japanese adults have a daily Mg deficit of ~50–150 mg, with particularly marked insufficiency among younger and middle-aged men.

Mg supplementation for quality-of-life improvement

Some individuals use Mg supplements for health promotion purposes, including prevention of nocturnal leg cramps, improvement of sleep quality, and relief of irritability or emotional tension. However, recent meta-analyses of Mg supplementation in healthy individuals (Mg ranging 232–439 mg/day) have failed to provide sufficient scientific evidence supporting significant benefits for sleep quality, muscle cramps, or anxiety reduction [55–57]. Accordingly, the appropriateness of Mg supplementation in these settings currently remains within the realm of empiric therapy.

Mg supplementation may also be relevant in dentistry. Approximately 60% of total body Mg is stored in bone and teeth, where it contributes alongside Ca and phosphate to enamel formation and maintenance of bone density. Even when Ca and vitamin D intake is adequate, insufficient Mg may impair normal bone formation. Mg also plays important roles in alveolar bone and gingival metabolism. Mg deficiency may promote alveolar bone resorption and gingival microcirculatory dysfunction, thereby contributing to periodontal disease. Indeed, epidemiological data from the U.S. National Health and Nutrition Examination Survey (NHANES) have demonstrated a positive association between Mg deficiency and periodontal disease [58]. Although current evidence does not support Mg supplementation as a substitute for standard periodontal therapy, adequate dietary intake or supplementation may contribute to prevention [59].

Conclusion

Both experimental and clinical evidence demonstrate that Mg deficiency contributes to BP elevation and that Mg supplementation exerts antihypertensive effects, suggesting that Mg plays an important role in the pathophysiology of BP regulation. However, there is currently no clinical evidence demonstrating that Mg-induced BP reduction could be extrapolated into the improvement of cardiovascular hard-endpoint including death, MACE and/or primary renal primary endpoints.

Nevertheless, recent advances in understanding renal tubular Na, K, and Mg transport systems have gained new insights into the Mg-related salt-sensitive hypertension. Although the precise mechanisms are multifactorial and yet largely unknown, the kidney can be regarded as the central integrator linking Mg homeostasis to BP regulation.

Acknowledgements The author gratefully acknowledges the valuable advice and insightful comments provided by Yoshindo Kawaguchi, MD, PhD, and Kuninobu Yokota, MD, PhD (Jikei University School of Medicine, Tokyo, Japan), as well as Isamu Taniguchi, DDS, PhD, and Jun Kuriyama, DDS (Taniguchi Dental Clinic, Kashiwa, Chiba, Japan), during the preparation of this manuscript.

Compliance with ethical standards

Conflict of interest The author declares no competing interests.

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